

Practice Problems: Linear Algebra

ECON 441: Introduction to Mathematical Economics

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Exercise 4.2

1. Given $A = \begin{bmatrix} 7 & -1 \\ 6 & 9 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 4 \\ 3 & -2 \end{bmatrix}$, and $C = \begin{bmatrix} 8 & 3 \\ 6 & 1 \end{bmatrix}$, find:

(a) $A + B$

(c) $3A$

(b) $C - A$

(d) $4B + 2C$

2. Given $A = \begin{bmatrix} 2 & 8 \\ 3 & 0 \\ 5 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 3 & 8 \end{bmatrix}$, and $C = \begin{bmatrix} 7 & 2 \\ 6 & 3 \end{bmatrix}$:

(a) Is AB defined? Calculate AB . Can you calculate BA ? Why?

(b) Is BC defined? Calculate BC . Is CB defined? If so, calculate CB . Is it true that $BC = CB$.

4. Find the product matrices in the following (in each case, append beneath every matrix a dimension indicator):

(a) $\begin{bmatrix} 0 & 2 & 0 \\ 3 & 0 & 4 \\ 2 & 3 & 0 \end{bmatrix} \begin{bmatrix} 8 & 0 \\ 0 & 1 \\ 3 & 5 \end{bmatrix}$

(c) $\begin{bmatrix} 3 & 5 & 0 \\ 4 & 2 & -7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

(b) $\begin{bmatrix} 6 & 5 & -1 \\ 1 & 0 & 4 \end{bmatrix} \begin{bmatrix} 4 & -1 \\ 5 & 2 \\ 0 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} 7 & 0 \\ 0 & 2 \\ 1 & 4 \end{bmatrix}$

Exercise 4.4

5. (e) Find (i) $C = AB$, and (ii) $D = BA$, if

$$A = \begin{bmatrix} -2 \\ 4 \\ 7 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 6 & -2 \end{bmatrix}$$

7. If the matrix A in Example 5 had all its four elements nonzero, would $x^T A x$ still give a weighted sum of squares? Would the associative law still apply?

Exercise 4.5

1. Given $A = \begin{bmatrix} -1 & 5 & 7 \\ 0 & -2 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 9 \\ 6 \\ 0 \end{bmatrix}$, and $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$:

Calculate: (a) AI (b) IA (c) Ix (d) $x^T I$

Indicate the dimension of the identity matrix used in each case.

4. Show that the diagonal matrix

$$\begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}$$

can be idempotent only if each diagonal element is either 1 or 0. How many different numerical idempotent diagonal matrices of dimension $n \times n$ can be constructed altogether from such a matrix?

Exercise 4.6

2. Given $A = \begin{bmatrix} 0 & 4 \\ -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -8 \\ 0 & 1 \end{bmatrix}$, and $C = \begin{bmatrix} 1 & 0 & 9 \\ 6 & 1 & 1 \end{bmatrix}$, verify that

(a) $(A + B)^T = A^T + B^T$

(b) $(AC)^T = C^T A^T$

6. Let $A = I - X(X^T X)^{-1} X^T$.

(a) Must A be square? Must $(X^T X)$ be square? Must X be square?

(b) Show that matrix A is idempotent.

Exercise 5.1

3. Are the rows linearly independent in each of the following?

$$(a) \begin{bmatrix} 24 & 8 \\ 9 & -3 \end{bmatrix}$$

$$(c) \begin{bmatrix} 0 & 4 \\ 3 & 2 \end{bmatrix}$$

$$(b) \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$(d) \begin{bmatrix} -1 & 5 \\ 2 & -10 \end{bmatrix}$$

4. Check whether the columns of each matrix in Prob. 3 are also linearly independent. Do you get the same answer as for row independence?

Exercise 5.2

1. Evaluate the following determinants:

$$(c) \begin{vmatrix} 4 & 0 & 2 \\ 6 & 0 & 3 \\ 8 & 2 & 3 \end{vmatrix}$$

$$(e) \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

$$(f) \begin{vmatrix} x & 5 & 0 \\ 3 & y & 2 \\ 9 & -1 & 8 \end{vmatrix}$$

2. Determine the signs to be attached to the relevant minors in order to get the following cofactors of a determinant: $|C_{13}|$, $|C_{23}|$, $|C_{33}|$, $|C_{41}|$, and $|C_{34}|$.

$$3. \text{ Given } \begin{vmatrix} a & b & c \\ d & e & f \\ g & b & i \end{vmatrix}, \text{ find the minors and cofactors of the elements } a, b \text{ and } f.$$

6. Find the minors and cofactors of the third row, given

$$A = \begin{bmatrix} 9 & 11 & 4 \\ 3 & 2 & 7 \\ 6 & 10 & 4 \end{bmatrix}$$

Exercise 5.3

$$1. \text{ Use the determinant } \begin{vmatrix} 4 & 0 & -1 \\ 2 & 1 & -7 \\ 3 & 3 & 9 \end{vmatrix} \text{ to verify the first four properties of determinants.}$$

4. Show that when all the elements of an n th-order determinant $|A|$ are multiplied by a number k , the result will be $k^n|A|$.
5. Calculate the determinant for the following matrices. Comment on whether the matrices are nonsingular and the rank of each matrix.

$$(a) \begin{bmatrix} 4 & 0 & 1 \\ 19 & 1 & -3 \\ 7 & 1 & 0 \end{bmatrix}$$

$$(b) \begin{bmatrix} 4 & -2 & 1 \\ -5 & 6 & 0 \\ 7 & 0 & 3 \end{bmatrix}$$

$$(c) \begin{bmatrix} 7 & -1 & 0 \\ 1 & 1 & 4 \\ 13 & -3 & -4 \end{bmatrix}$$

$$(d) \begin{bmatrix} -4 & 9 & 5 \\ 3 & 0 & 1 \\ 10 & 8 & 6 \end{bmatrix}$$

8. Comment on the validity of the following statements:

- (a) Given any matrix A , we can always derive from it a transpose and a determinant.
- (b) Multiplying each element of an $n \times n$ determinant by 2 will double the value of that determinant.
- (c) If a square matrix A vanishes, then we can be sure that the equation system $Ax = d$ is nonsingular.

Exercise 5.4

2. Find the inverse of each of the following matrices:

$$(a) A = \begin{bmatrix} 5 & 2 \\ 0 & 1 \end{bmatrix}$$

$$(c) C = \begin{bmatrix} 3 & 7 \\ 3 & -1 \end{bmatrix}$$

$$(b) B = \begin{bmatrix} -1 & 0 \\ 9 & 2 \end{bmatrix}$$

$$(d) D = \begin{bmatrix} 7 & 6 \\ 0 & 3 \end{bmatrix}$$

- 3. (a) Drawing on your answers to Prob. 2, formulate a two-step rule for finding the adjoint of a given 2×2 matrix A : In the first step, indicate what should be done to the two diagonal elements of A in order to get the diagonal elements of $\text{adj}A$; in the second step, indicate what should be done to the two off-diagonal elements of A . (Warning: This rule applies only to 2×2 matrices.)
 - (b) Add a third step which, in conjunction with the previous two steps, yields the 2×2 inverse matrix A^{-1} .
4. Find the inverse of each of the following matrices:

$$(a) \ E = \begin{bmatrix} 4 & -2 & 1 \\ 7 & 3 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

$$(b) \ F = \begin{bmatrix} 1 & -1 & 2 \\ 1 & 0 & 3 \\ 4 & 0 & 2 \end{bmatrix}$$

$$(c) \ G = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$(d) \ H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

7. Is it possible for a matrix to be its own inverse?

Exercise 5.4

6. Solve the system $Ax = d$ by matrix inversion, where

$$(a) \ 4x + 3y = 28$$

$$2x + 5y = 42$$

$$(b) \ 4x_1 + x_2 - 5x_3 = 8$$

$$-2x_1 + 3x_2 + x_3 = 12$$

$$3x_1 - x_2 + 4x_3 = 5$$

Exercise 5.5

1. Use Cramer's rule to solve the following equation systems:

$$(a) \ 3x_1 - 2x_2 = 6$$

$$2x_1 + x_2 = 11$$

$$(b) \ -x_1 + 3x_2 = -3$$

$$4x_1 - x_2 = 12$$

$$(c) \ 8x_1 - 7x_2 = 9$$

$$x_1 + x_2 = 3$$

$$(d) \ 5x_1 + 9x_2 = 14$$

$$7x_1 - 3x_2 = 4$$

2. For each of the equation systems in Prob. 1, find the inverse of the coefficient matrix, and get the solution by the formula $x^* = A^{-1}d$.

3. Use Cramer's rule to solve the following equation systems:

$$(a) \ 8x_1 - x_2 = 16$$

$$2x_2 + 5x_3 = 5$$

$$2x_1 + 3x_3 = 7$$

$$(d) \ -x + y + z = a$$

$$x - y + z = b$$

$$x + y - z = c$$